

CONSTRUCTION AND PROJECT MANAGEMENT - I

Section A - Lecture 08

SCHEDULING FOR REPETITIVE PROJECTS

Term-Final Question-to-Slide Mapping | 2016-2024

Line of Balance Method (LOB)

- Line of balance (LOB) method is a technique used in the planning and scheduling of a construction project which consists of a number of identical units with the various construction operations being repeated from unit to unit
- A technique for scheduling linear projects that involve a recognizable number of repetitive activities
- Examples of linear projects: highways, pipelines, high-rise buildings

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LOB definition and applications

Terminology used in LOB

- Specialist Gangs:
 - Each operation is to be carried out by a specialist gang. Example, a concreting gang will only do concreting.
 - Each specialist gang consist of a number of teams (or crews).
- Example: There are 4 Teams in a Specialist Gang A
 - (identified as team A1, team A2, A3 and A4).
 - Team A1 will carry out Operation A in Unit 1
 - Team A2 will carry out Operation A in Unit 2
 - Team A3 will carry out Operation A in Unit 3
 - Team A4 will carry out Operation A in Unit 4

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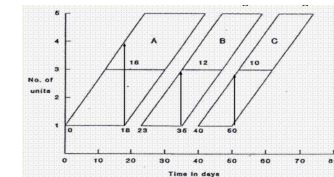
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Specialist gangs and teams

The theoretical schedule

Step 4: Identify the required number of team to meet deadline



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Theoretical repetitive schedule

Prepared from: Lecture 08, *Scheduling for Repetitive Projects*, and the Construction and Project Management - I question bank (2016-2024). The mapping follows topic content rather than the changing A/B section labels.

Important: The lecture contains a worked jetty example. The recurring pipeline questions use the same LOB/LSM method but different project data, so they are classified as **Direct**, not Exact.

1. Executive Summary

12

Mapped questions

Across all 9 years

146

Mapped marks

2016-2024 total

6 Exact + 6 Direct

Match profile

No partial-only items

8

Highest-priority slide

LOB definition

Core finding: This is a high-frequency examination lecture. Every year from 2016 to 2024 contains at least one related question. The dominant pattern is the five-operation pipeline LSM problem, repeated in six consecutive examinations from 2017 through 2022.

Year	Original section and question(s)	Marks	Coverage focus
2016	A 4(a), A 4(c)	25	Repetitive construction; LOB terminology
2017	A 1(c)	15	Pipeline LSM numerical
2018	B 6(b), B 6(c)	20	Linear-project method; pipeline LSM
2019	B 8(c)	15	Pipeline LSM numerical
2020	B 6(b), B 6(c)	20	Linear-project method; pipeline LSM
2021	B 8(c)	15	Pipeline LSM with 2-day buffer
2022	B 8(c)	16	Pipeline LSM with 2-day buffer
2023	B 7(a)	10	Repetitive construction
2024	B 7(a)	10	Specialist gang, team size, build rate

Match rules: Exact = the lecture directly contains the requested definition or terminology. Direct = the lecture teaches the required method and an analogous worked example, but the exact examination dataset is different.

2. Detailed Question-to-Slide Mapping

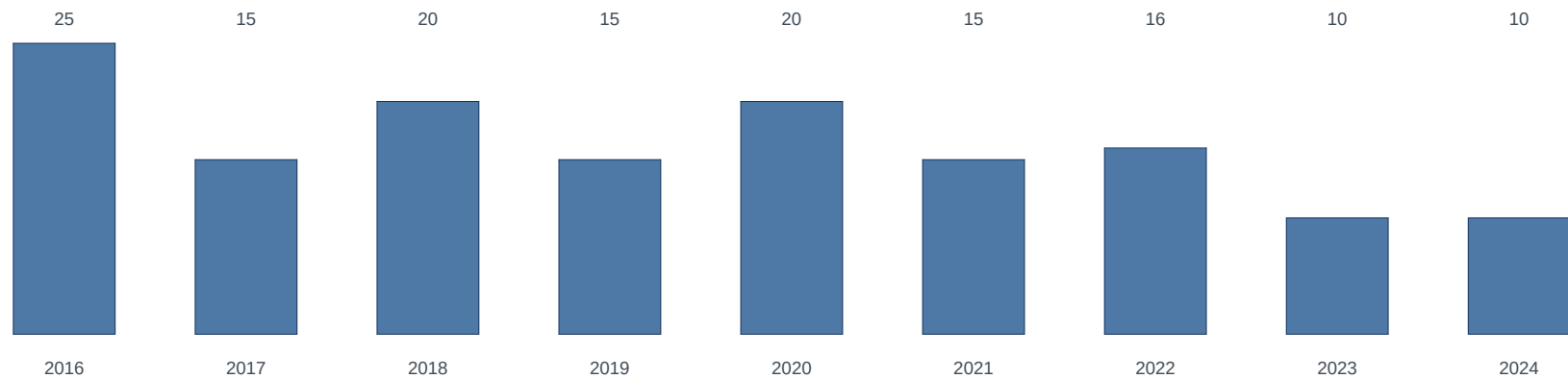
Year	Sec.	Q.	Marks	QB page	Question focus	Lecture slide / PDF page	Match	Coverage note
2016	A	4(a)	10	7	Define repetitive construction and write its characteristics.	L4-6 / P5-7	Exact	The lecture directly defines repetition of operations, crews, equipment and resources across units, with wall-panel and road examples.
2016	A	4(c)	15	7	Short notes: specialist gang, optimum team size and target build rate.	L9-11 / P10-12	Exact	All three terms are defined separately, including gang-team relationships, $g = n \times Q$, and the meaning of target build rate R.
2017	A	1(c)	15	4	Five-operation 1200 m pipeline; draw an LSM diagram with a one-day buffer.	L7, L12-20 / P8, P13-21	Direct	The lecture gives the LOB/LSM definition, chart interpretation, scheduling method and a complete repetitive-project example, but not this exact pipeline dataset.
2018	B	6(b)	8	13	Which technique is used for planning and scheduling a linear project? Briefly explain.	L3, L7 / P4, P8	Exact	Slides identify LOB and LSM as repetitive-project methods and define LOB for linear projects such as highways and pipelines.
2018	B	6(c)	12	13-14	Five-operation 1000 m pipeline; draw an LSM diagram with a one-day buffer.	L7, L12-20 / P8, P13-21	Direct	The procedure and graphical logic are fully covered; the examination dataset must be worked separately.
2019	B	8(c)	15	11	Five-operation 1000 m pipeline; explain and draw an LSM diagram with a one-day buffer.	L7, L12-20 / P8, P13-21	Direct	Same recurring LSM application; lecture provides method, chart reading, rate concept and an analogous worked example.
2020	B	6(b)	8	13-14	Which technique is used for planning and scheduling a linear project? Briefly explain.	L3, L7 / P4, P8	Exact	Directly supported by the planning-method classification and LOB definition slides.
2020	B	6(c)	12	14	Five-operation 1000 m pipeline; draw and explain an LSM diagram with a one-day buffer.	L7, L12-20 / P8, P13-21	Direct	The lecture explains how to construct and interpret the repetitive schedule; exact numerical solution is not printed.
2021	B	8(c)	15	17	Five-operation 1000 m pipeline; draw an LSM diagram with a two-day buffer.	L7, L12-20 / P8, P13-21	Direct	Same LSM framework with a different stated buffer. Use the lecture method and preserve the two-day separation.
2022	B	8(c)	16	19	Five-operation 1000 m pipeline; draw and explain an LSM diagram with a two-day buffer.	L7, L12-20 / P8, P13-21	Direct	The complete process is supported by the theoretical chart and worked jetty example; the pipeline values are not duplicated.
2023	B	7(a)	10	21	Define repetitive construction and write its characteristics.	L4-6 / P5-7	Exact	A near-repeat of 2016 Q4(a); the required characteristics are stated and illustrated.
2024	B	7(a)	10	25	Write notes on specialist gang, optimum team size and target build rate.	L9-11 / P10-12	Exact	A near-repeat of 2016 Q4(c), with the same three lecture terms.

Lecture slide number means the number printed on the slide. PDF page means the file page. Example: L7 / P8 is lecture slide 7 on PDF page 8.

3. Repeated Question Patterns

Pattern	Years	Frequency	Marks	Primary slides
Pipeline LSM numerical	2017, 2018, 2019, 2020, 2021, 2022	6	85	L7, L12-20
Repetitive construction and characteristics	2016, 2023	2	20	L4-6
Specialist gang, optimum team size, target build rate	2016, 2024	2	25	L9-11
Technique for a linear project	2018, 2020	2	16	L3, L7

Mapped marks by examination year



Interpretation: The lecture appears every year. The largest single-year contribution is 25 marks in 2016, while the 2017-2022 period is dominated by the recurring pipeline LSM problem.

4. Slide-to-Question Reverse Index

Slide / page	Lecture content	Question-bank use
L2 / P3	Planning versus scheduling	Supports explanation of how a schedule translates a plan into time, sequence and dependency.
L3 / P4	CPM, PERT, LOB and LSM classification	2018 B 6(b); 2020 B 6(b).
L4-6 / P5-7	Characteristics of repetitive projects	2016 A 4(a); 2023 B 7(a).
L7 / P8	LOB definition and linear-project examples	All six pipeline LSM questions; also 2018 B 6(b) and 2020 B 6(b).
L8 / P9	Objective of LOB	Useful supporting material for theory answers and numerical interpretation.
L9 / P10	Specialist gangs and teams	2016 A 4(c); 2024 B 7(a).
L10 / P11	Optimum team size and actual gang size	2016 A 4(c); 2024 B 7(a).
L11 / P12	Target build rate	2016 A 4(c); 2024 B 7(a); supports recurring numerical questions.
L12-14 / P13-15	LOB chart interpretation and rate calculation	All pipeline LSM questions from 2017-2022.
L15 / P16	Four-stage method to develop an LOB schedule	All pipeline LSM questions from 2017-2022.
L16-20 / P17-21	Worked five-unit jetty example	Closest worked model for all pipeline LSM questions.

Most important cluster: L7 and L12-20. Together they cover the definition, chart reading, rate concept, construction method and worked example required for the repeated numerical question.

5. Exam-Focused Answer Construction

A. Linear-project scheduling technique (2018, 2020)

1	Name the method: Line of Balance (LOB) / Linear Scheduling Method (LSM).
2	Define it: a planning and scheduling technique for projects made of identical units or recognizable repetitive activities.
3	Give examples: highways, pipelines and high-rise buildings.
4	Add the objective: balance resources and synchronize crews so work remains continuous and non-interrupted.

B. Repetitive construction characteristics (2016, 2023)

- The work process completed in one unit is repeated in another unit.
- A crew that performs an activity in one unit can repeat it in the next unit.
- Equipment and other resources are also reused from unit to unit.
- Each unit should contain a set of well-defined operations or stages.
- Use one illustration: repeated wall panels or road segments.

C. Short-note terminology (2016, 2024)

Specialist gang	A gang dedicated to one operation. It may contain several teams or crews working on different units.
Optimum team size (Q)	The number of workers at which a specialist team achieves its highest productivity.
Actual gang size (g)	An integer multiple of the optimum team size: $g = n \times Q$.
Target build rate (R)	The completion rate of units after the first unit has been completed, commonly expressed in units per week.

6. Numerical LSM Question Strategy and Revision Plan

A. Source-supported solution framework for the recurring pipeline problem

1	Identify the operations associated with one repeating unit or segment.
2	Construct the logical sequence between operations for one unit.
3	Develop the theoretical LOB/LSM chart using the stated productivity rates.
4	Insert the examination buffer exactly as stated: one day in 2017-2020 and two days in 2021-2022.
5	Check that successive crew lines do not conflict and that work remains continuous.
6	Label axes, operations, units/length, start and finish positions, and explain the schedule.

Limitation: The uploaded lecture does not print the exact 1000 m or 1200 m pipeline solution. It provides the method and an analogous five-unit jetty example. Therefore, the six pipeline mappings are correctly marked Direct.

B. High-priority revision plan

Priority	Slides	Revision task	Reason
1	L7, L12-15	Master the LOB/LSM definition, chart interpretation, rate and four-stage construction method.	Highest - 6 numerical repeats
2	L16-20	Reproduce the jetty example without looking at the slides.	Model for pipeline problems
3	L9-11	Memorize specialist gang, Q , $g = n \times Q$ and target build rate.	Repeated in 2016 and 2024
4	L4-6	Learn the four repetitive-project characteristics plus one example.	Repeated in 2016 and 2023
5	L3, L7-8	Prepare an 8-mark theory answer on linear-project scheduling.	Repeated in 2018 and 2020

Overall verdict: Lecture 08 is a very high-yield topic. It generated 12 mapped questions and 146 marks across 2016-2024, with no examination year left uncovered.